

Session Objectives

- **Describe** the main features for the different types of lipoprotein disorders and explain their role in disease
- **Identify** the main features for primary causes of dyslipidemia
- **Examine** the main features for secondary causes of dyslipidemia
- **Correlate** the presenting signs and symptoms with the different types of dyslipidemia disorders
- **Explain** the rationale for the different preventative strategies for dyslipidemia disorders and formulate a treatment plan based on the latest guidelines

Dyslipidemia – Clinical Significance

- Dyslipidemias contribute to the development of **atherosclerosis**
 - Coronary artery disease
 - Peripheral vascular disease
 - Cerebrovascular accidents
- **Severe hypertriglyceridemia may lead to acute pancreatitis**
 - Critical element in non-alcoholic fatty liver disease

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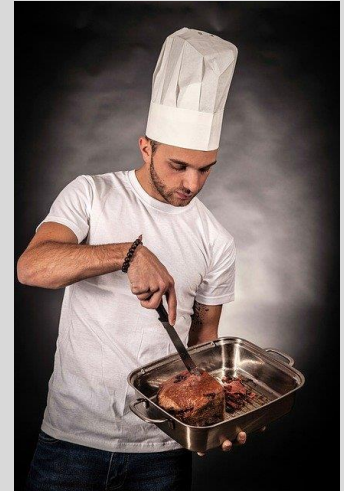
Source: Disorders of Lipoprotein Metabolism: Harrison's Principles of Internal Medicine, 2025, 22nd edition, Chapter 419

Lipoproteins and Atherosclerosis

- **Atherosclerosis** – Very common disease of the arteries caused by buildup of cholesterol plaques
 - Macrophage and LDL accumulation eventually lead to formation of complex atheromas
 - Complications: Ischemia, thrombus, emboli and aneurysms
 - Symptoms: Angina, claudication or asymptomatic

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Risk Factors for Developing Atherosclerosis

- Age – increases with each decade of life
- Male gender
- Family history:
 - Hypertension
 - Diabetes
 - Familial hypercholesterolemia
- **Cigarette smoking**
- Hypertension - both systolic and diastolic
- Hyperlipidemia - primarily elevated LDL and low HDL
- Diabetes
- Metabolic syndrome

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Metabolic syndrome

- Group of risk factors
- Defined by **three or more of the following** five abnormalities:
 - **Fasting** blood glucose ≥ 100 mg/dl
 - Blood pressure of at least **130/85 mm Hg**
 - Triglycerides ≥ 150 mg/dl
 - HDL level of **< 40 mg/dl** in men or **< 50 mg/dl** in women
 - Waist circumference **> 102 cm (40 in)** in men or **> 88 cm (35 in)** in women

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Primary causes of dyslipidemia

- **Familial hypercholesterolemia**
 - Genetic defect: **LDL receptor defect**
 - **Diminished LDL clearance**
 - Inheritance: Co-dominant or complex with multiple gene involvement
 - Prevalence:
 - Heterozygotes 1/200
 - Homozygotes 1/250,000 - 1/1 million
 - Heterozygotes: TC 250-500 mg/dl
 - Premature CAD (Ages 30-50) – 5% of MI's <60 yr
 - Homozygotes: TC > 500 mg/dl
 - Premature CAD (Before age 18)

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Source: Dyslipidemia – Endocrine and Metabolic Disorders - Merck Manuals Professional Edition
– Last full review/revision August 2026 by Michael H. Davidson, MD

Primary causes of dyslipidemia

- **Familial combined hyperlipidemia**
 - Genetic defect: **Unknown, possible genetic multiple defects and mechanisms**
 - Inheritance: Autosomal dominant
 - Prevalence: 1/50 to 1/100
 - Premature CAD
 - 15% of MI's in people < 60 yr
 - **Disproportionately elevated apoB**
 - TC 250-500 mg/dl
 - TG 250-750 mg/dl

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Primary causes of dyslipidemia

- **Familial hypertriglyceridemia**
 - Genetic defect: Unknown, possible genetic multiple defects and mechanisms
 - Inheritance: Autosomal dominant
 - Prevalence: 1/500
 - Usually no symptoms or findings
 - Absence of secondary causes
 - **TG: 200-500 mg/dl**

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Secondary causes of dyslipidemia

- Other common secondary causes include:
 - Endocrine disorders:
 - **Diabetes mellitus**, Insulin resistance, Cushing disease, **Hypothyroidism**,
 - Hepatic disorders:
 - **Primary biliary cirrhosis**, Other cholestatic liver diseases
 - Renal disorders:
 - Chronic kidney disease, **Nephrotic syndrome**
 - Medications:
 - **Thiazides**, **β -blockers**, Retinoids, Highly active antiretroviral agents, Estrogen and progestins, Glucocorticoids, Cyclosporine and Tacrolimus
 - Other: Diet, **Alcohol overuse**, **Obesity**

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Source: Dyslipidemia – Endocrine and Metabolic Disorders - Merck Manuals Professional Edition
– Last full review/revision August 2026 by Michael H. Davidson, MD

Secondary causes of dyslipidemias

- **Excessive hepatic production of VLDL**
 - Characterized by elevated fasting TG, Low HDL, variable levels of LDL
- Common causes:
 - **High carbohydrate diet**
 - Obesity and insulin resistance
 - **More free fatty acids** are delivered from the adipose tissue to the liver
 - Increased insulin levels promote increased fatty acid synthesis
 - Nephrotic syndrome
 - Cushing's syndrome

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Secondary causes of dyslipidemias

- **Alcohol overuse**
 - Causes **excessive hepatic production of VLDL**
 - **Inhibits hepatic oxidation** of free fatty acids
 - **Promotes hepatic TG synthesis** and VLDL secretion
 - Raises plasma levels of HDL
 - Consider in persons with combination of **elevated TG and HDL**

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Secondary causes of dyslipidemias

- Impaired Lipolysis of TG-rich lipoprotein's (TRL)
 - Due to decreased LPL (Lipoprotein lipase activity)
 - Characterized by elevated fasting TG, Low HDL usually without elevations of LDL or apoB
- Common causes:
 - **Obesity and insulin resistance**

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Secondary causes of dyslipidemias

- **Impaired Hepatic Uptake of Lipoproteins containing ApoB**
 - Reduced LDL receptor activity
- Characterized by elevated LDL
- Common causes:
 - **Hypothyroidism**
 - Increased IDL and mild elevation of TG
 - **Chronic kidney disease**
 - Mild hypertriglyceridemia (<300 mg/dl)

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Secondary causes of dyslipidemias

- **Reduced HDL levels**
 - Combination of genetic predisposition and secondary factors
 - **Independent predictor for increased cardiovascular risk**
 - HDL metabolism is influenced by TRL, insulin resistance and inflammation
- Secondary causes of low levels of HDL cholesterol include:
 - Cigarette smoking, Anabolic steroids, HIV infection, Nephrotic syndrome

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Secondary causes of dyslipidemias

- **Liver disorders**
 - Hepatitis
 - Infection, drugs or alcohol are associated with increased VLDL synthesis and mild to moderate hypertriglyceridemia
 - Cholestasis is associated with hypercholesterolemia which can be severe
- **Drugs**
 - Beta blockers – Reduced HDL, elevated VLDL
 - Thiazides – Elevated LDL
 - Estrogen – Elevated HDL, elevated VLDL & TG

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Signs and Symptoms

- Dyslipidemia itself is usually **asymptomatic**
- **BUT**
 - Leads to **symptomatic vascular disease** including:
 - Coronary artery disease (CAD)
 - Stroke
 - Peripheral arterial disease
- Extremely high triglyceride levels can give a lactescent appearance to blood plasma.
 - **Symptoms:** Paresthesias, dyspnea and confusion

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Signs and Symptoms

- High levels of TGs (**> 1000 mg/dl**) can cause acute pancreatitis
- Severe elevations of TGs can present with:
 - Eruptive xanthomas over the trunk, back, elbows, buttocks, knees, hands and feet
- Severe hypertriglyceridemia (**>2000 mg/dl**) can present with:
 - Lipemia retinalis

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Source: S. Kang, M. Amagai, A.L. Bruckner, A.H. Enk, D.J. Margolis, A.J. McMichael, J.S. Orringer: Fitzpatrick's Dermatology, Ninth Edition Copyright © McGraw-Hill Education. All rights reserved.

Source: Dyslipidemia – Endocrine and Metabolic Disorders - Merck Manuals Professional Edition – Last full review/revision August 2026 by Michael H. Davidson, MD

Dyslipidemia – Clinical Manifestation

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- Dyslipidemias may *manifest* as one of more of the following:
 - Elevated total plasma cholesterol
 - Elevated low-density lipoproteins (LDL)
 - Elevated triglycerides (TGs)
 - Decreased high-density lipoprotein (HDL)



Source: S. Kang, M. Amagai, A.L. Bruckner, A.H. Enk, D.J. Margolis, A.J. McMichael, J.S. Orringer: Fitzpatrick's Dermatology, Ninth Edition Copyright © McGraw-Hill Education. All rights reserved.

Diagnosis

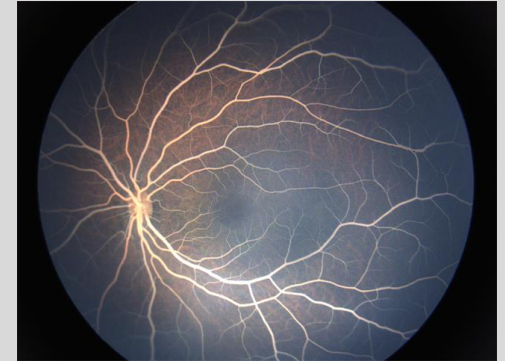
- Dyslipidemia is diagnosed by measuring serum lipids
- Serum lipid profile
 - Total cholesterol
 - Triglycerides
 - HDL cholesterol
 - Calculated LDL cholesterol and VLDL cholesterol

$$*TC = VLDL + LDL + HDL$$

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Serum Lipid Profile

- TC, TGs, and HDL cholesterol are measured directly
 - TC values can vary by 10% and TGs by up to 25% day-to-day even in the absence of a disorder
- Patients should have **all lipids measured while fasting (usually for 12 h)** for maximum accuracy and consistency
- Testing should be **postponed** until after resolution of acute illness since:
 - TG and lipoprotein levels **increase**, and cholesterol levels **decrease** in inflammatory states

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Screening for Dyslipidemia

- **Universal screening** using a ***fasting*** lipid profile should be done:
 - No risk factors - Once for all children between age 9 and 11
 - or at 2 to 8 if children have a family history of severe hyperlipidemia or premature CAD
 - **Adults are screened at age 20 yr** and every 5 years thereafter
 - Assessment for cardiovascular risk factors should accompany the lipid measurements
 - *A definite age to discontinue screening has not been determined*



Screening for secondary causes of dyslipidemia

- Patients with newly diagnosed dyslipidemia **and/or** when a component of the lipid profile has inexplicably **changed for the worse**
 - Fasting glucose
 - Liver enzymes
 - Creatinine
 - Thyroid-stimulating hormone (TSH)
 - Urinary protein
- Once secondary causes ruled out attempt to determine primary cause of dyslipidemia

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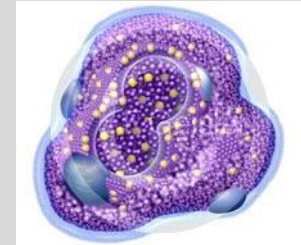


Dyslipidemias - Diagnosis of Primary Cause

- **Chylomicronemia:** Fasting plasma TG level $> 1,000$ mg/dl
 - Familial chylomicronemia syndrome: **Cholesterol to TG ratio > 10**
 - At risk for acute pancreatitis
- **Familial hypercholesterolemia:**
 - LDL levels are elevated in the absence of hypertriglyceridemia
 - Clinical diagnosis

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Dyslipidemias – Diagnosis of Primary Cause

- **Combined hyperlipidemia**
 - Elevation of TG, VLDL and LDL
 - Ascertain the VLDL/TG ratio in plasma or direct LDL to determine if hyperlipidemia is due to:
 - Accumulation of lipoprotein remnants (i.e. Familial dysbetalipoproteinemia) or
 - An increase in both LDL and VLDL
 - **Measurement of apoB levels** can help identify patients with Familial combined hyperlipidemia

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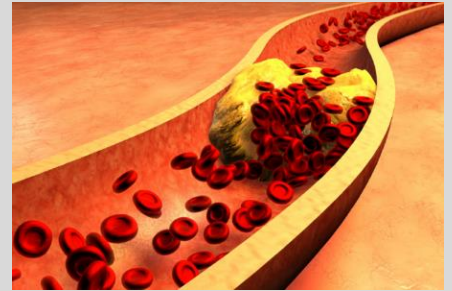
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ACC/AHA Guidelines 2026

- Main indication for dyslipidemia treatment is prevention of atherosclerotic cardiovascular disease (ASCVD)
 - Acute coronary syndromes
 - Stroke, transient ischemic attack
 - Peripheral vascular disease

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Source: Dyslipidemia – Endocrine and Metabolic Disorders - Merck Manuals Professional Edition
– Last full review/revision August 2026 by Michael H. Davidson, MD

ACC/AHA Guidelines 2026 – Key Points

- Treat dyslipidemia earlier to reduce lifelong risk of prolonged exposure to atherogenic lipoprotein.
 - Lifestyle optimization should start in youth
 - Consider pharmacotherapy in youth with familial hypercholesterolemia and in young adults with LDL-C > 2160 mg/dl or family history of premature ASCVD
- Use the AHA PREVENT risk calculator to determine 10-year risk of ASCVD

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ACC/AHA Guidelines 2026 – Key Points

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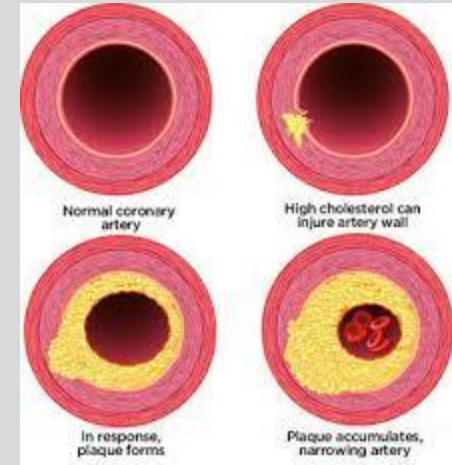
- In secondary prevention a goal of LDL-C <55 mg/dl and non-HDL-C <85 mg/dl is recommended for those at very high risk of ASCVD events
- Patient with persistently elevated triglycerides statin therapy remains the foundation pharmacotherapy along with lifestyle intervention to reduce ASCVD risk.
- In individuals with TG levels of >1000 mg/dl treatment with TG-lowering therapies helps prevent pancreatitis

ACC/AHA Guidelines 2026

- AHA/ACC recommends treatment with a statin for 4 groups of patients:
 - Presence of clinical atherosclerotic cardiovascular disease
 - LDL cholesterol ≥ 190 mg/dl
 - Age 40 to 75 with diabetes and LDL cholesterol 70 to 189 mg/dl
 - Age 40 to 75, LDL cholesterol 70 to 189 mg/dl and estimated 10-yr ASCVD $\geq 7.5\%$

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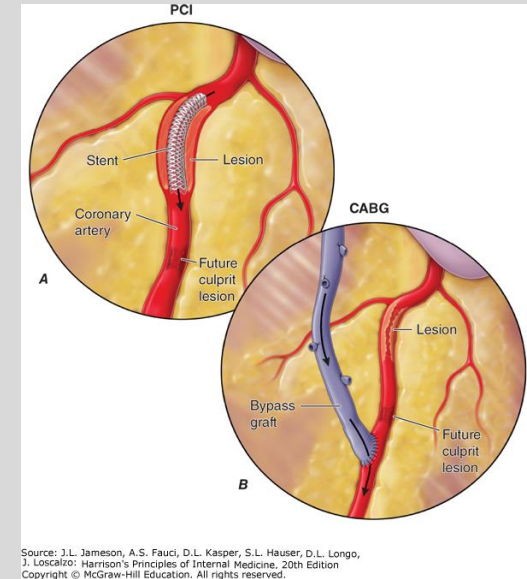


ACC/AHA Guidelines 2019

- Risk of ASCVD is estimated using a risk calculator that takes into account:
 - Sex
 - Age
 - Race
 - Total and HDL cholesterol
 - Systolic and diastolic BP
 - Use of antihypertensives or statins
 - Presence or absence of diabetes
 - **Smoker status**

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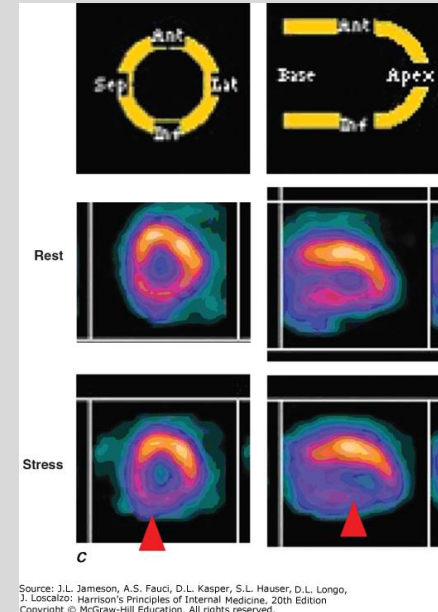
ACC/AHA Guidelines 2019

- Other factors to consider to determine need for a statin:
 - LDL cholesterol ≥ 160 mg/dl
 - Family history of premature CAD
 - Onset < 55 yr in male 1st degree relative
 - Onset < 65 yr in a female 1st degree relative
 - High sensitivity C-reactive protein ≥ 2 mg/L
 - Coronary artery calcium score ≥ 300 Agatson units (or > 75th percentile)
 - Ankle-brachial BP index < 0.9
 - Increased lifetime risk (ACC/AHA PREVENT risk calculator)

Source: Dyslipidemia – Endocrine and Metabolic Disorders - Merck Manuals Professional Edition
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Goals of Treatment of Lipoprotein Disorders

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- Major goals:
 - Prevention of CVD and related cardiovascular events
 - Prevention of acute pancreatitis in patients with severe hypertriglyceridemia



Source: Disorders of Lipoprotein Metabolism: Harrison's Principles of Internal Medicine, 2025, 22nd edition, Chapter 407

Treatment Plan for Lipoprotein Disorders

- Risk assessment for ASCVD
- Lifestyle changes- 1st line
 - Exercise
 - Dietary modification
- Elevated LDL: statins, bile acid sequestrants, ezetimibe, niacin, bempedoic acid, PSK9 monoclonal antibodies
- Elevated TG: Niacin, fibrates, omega-3-fatty acids

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Management for Hypercholesterolemia

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- Lifestyle:
 - If obese - Reduce weight to ideal level
 - Aerobic exercise
- Diet
 - Reduce intake of: Saturated fats, Trans fats, Cholesterol
 - Increase intake of: Fiber, Complex carbohydrates



Source: Dyslipidemia – Endocrine and Metabolic Disorders - Merck Manuals Professional Edition
– Last full review/revision August 2026 by Michael H. Davidson, MD

Management of Hypercholesterolemia

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- **Statins are the treatment of choice for reduction of LDL cholesterol**
 - Reduction of cardiovascular risk is well supported by clinical trial data
 - Statins = Hydroxymethylglutaryl Coenzyme A (HMG-CoA) Reductase Inhibitors
 - Statins work by inhibiting the rate-limiting enzyme in cholesterol biosynthesis
 - **Significantly reduce LDL and have a modest HDL – raising effect**



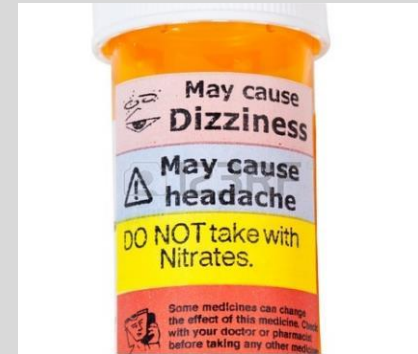
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Management of Hypercholesterolemia

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- Statins are mainly well tolerated
 - Side effects:
 - Dyspepsia
 - Headaches
 - Fatigue
 - **Muscle or joint pain**
 - Elevation of liver transaminase levels
 - Need to closely monitor at least every 3 months



Source: Disorders of Lipoprotein Metabolism: Harrison's Principles of Internal Medicine, 2022, 21th edition, Chapter 407

Management of Hypertriglyceridemia

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- Lifestyle: (**First Line**)
 - If obese - Reduce weight to ideal level
 - Aerobic exercise
- Dietary:
 - **Reduce intake of:**
 - Alcohol or preferably eliminate
 - Simple carbohydrates
 - Increase intake of:
 - Omega-3-fatty acids (Marine fish)
- Diabetics:
 - Tight control of glucose levels



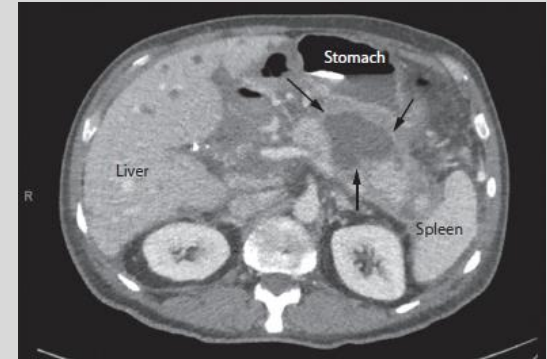
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Management of Hypertriglyceridemia

- Failed lifestyle management
- Initiate medications for patients with persistent **TG levels > 500 mg/dl**
 - Fibric acid derivatives
 - Side effects: Myopathy, gallstones, creatinine elevations
 - Note: May raise LDL levels
 - Omega 3 Fatty Acids
 - Side effects: Dyspepsia
 - Note: May prolong bleeding time
 - Nicotinic Acid (Niacin)
 - Side effects: Dyspepsia
 - Note: May elevate uric acid and liver transaminase

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Source: Jesse B. Hall, Gregory A. Schmidt, John P. Kress: *Principles of Critical Care*, 4th Edition: www.accessmedicine.com
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Summary Key Points

- Elevated lipid levels are a **risk factor** for atherosclerosis
- Causes of dyslipidemia include **sedentary lifestyle** with **excessive dietary intake** and or genetic abnormalities
- Diagnose using **fasting** serum lipid profile
- Screening tests should be done at age **9 to 11**
 - **Age 2** if there is strong family history of severe hyperlipidemia or premature CAD
- **Screen adults every 5 years after age 20**

